Day 11: The strip on the line

Name _____ Date _____

I can find a fraction's other names on a number line, past 1 too, and say why the point stays put.

Warm-up.

1. 465 + 278 = ? _____	2. 802 − 357 = ? _____	3. 8 × 7 = ? _____

Example 1. The line from 0 to 1 is marked in fourths. A strip $\frac{1}{4}$ long lies on it from 0. Cut each fourth into 2 equal parts. What is the new name for the strip's end?

The new name: ___/___ Did the point move? _____

Up to 1. Use your number lines sheet. Count parts, not marks.

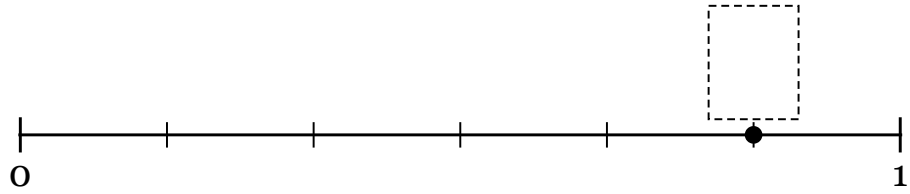
4. A line from 0 to 1 is in fourths. A point is at $\frac{2}{4}$. Cut each fourth into 2. New name? New name (line 2): ___/___
5. A line from 0 to 1 is in fifths. A point is at $\frac{4}{5}$. Cut each fifth into 2. New name?

   New name (line 3): ___/___
6. On the fifths line, a point is at 1. Name it in fifths, then in tenths. Line 3.

   In fifths: ___/___ In tenths: ___/___

Fix it. Ana is wrong. Say what went wrong. Then name her point.

7. Ana marked sixths on a line from 0 to 1. She counted every mark from the mark at 0 to her point: 1, 2, 3, 4, 5, 6. She named the point $\frac{6}{6}$.



What went wrong: _____ Her point is ___/___.

Past 1. Use lines 4 to 6. The marks are printed. To cut each part in 2, mark its middle.

Example 2. The line from 0 to 2 is marked in halves. A point is at $\frac{3}{2}$. Cut each half into 4 equal parts. What is the point's new name?

The new name: ___/___ Did the point move? _____

8. A line from 0 to 2 is in fifths. A point is at $\frac{7}{5}$. Cut each fifth into 2. New name?

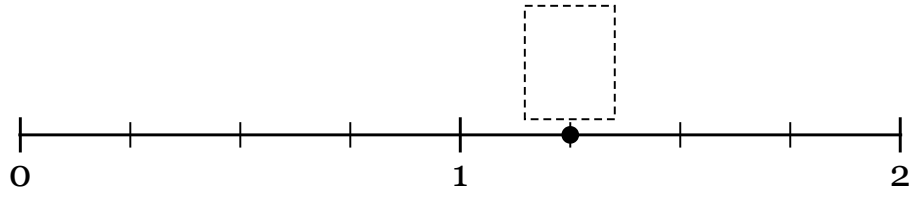
   New name (line 5): ___/___

9. A line from 0 to 2 is in fourths. A point is at 2. Name it in fourths. Then cut each fourth into 2 and name it again. Line 6.

   In fourths: ___/___ In eighths: ___/___

Fix it. Max is wrong. Say what went wrong. Then name his point.

10. Max's line from 0 to 2 is marked in fourths. His point is on the fifth mark after 0. He counted all 8 parts and named it $\frac{5}{8}$.



What went wrong: _____

His point is ___/___.

What changed? Each part was cut into ___, so there are ___ times as many parts and each part is smaller. The length is the same, so ___/___ = ___/___.

Challenge. Can $\frac{3}{8}$ be written in fourths? Why or why not?

_____

_____

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